

Perspective

The purpose and ubiquity of turnover

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SUMMARY

Turnover—constant component production and destruction—is ubiquitous in biology. Turnover occurs across organisms and scales, including for RNAs, proteins, membranes, macromolecular structures, organelles, cells, hair, feathers, nails, antlers, and teeth. For many systems, turnover might seem wasteful when degraded components are often fully functional. Some components turn over with shockingly high rates and others do not turn over at all, further making this process enigmatic. However, turnover can address fundamental problems by yielding powerful properties, including regeneration, rapid repair onset, clearance of unpredictable damage and errors, maintenance of low constitutive levels of disrepair, prevention of stable hazards, and transitions. I argue that trade-offs between turnover benefits and metabolic costs, combined with constraints on turnover, determine its presence and rates across distinct contexts. I suggest that the limits of turnover help explain aging and that turnover properties and the basis for its levels underlie this fundamental component of life.

INTRODUCTION

General principles in biology seek to broadly explain and predict system attributes. This article addresses the basis for and ubiquity of turnover in biology. Turnover here refers to the continuous production and degradation of system components, even when fully functional (Figure 1A). Turnover is everywhere and can be stunningly rapid. Recall a memory from weeks past and it flows through different molecules than before. Touch a familiar object from months ago and new skin cells made contact. Eat something you haven't had in weeks and new taste buds detected the flavor and the food will pass by entirely new cells in the intestine. Look at something you haven't seen in weeks and the light will pass through new cells in your cornea and be detected by new membranous discs in your photoreceptor neurons. Today, you will produce 250 billion red blood cells (RBCs) and replace most mRNA in your cells several times over. In some species, like planarians and hydra, all cells of the body turn over.

Human experience with the non-living can inform intuitive frameworks used for understanding biology, but these can be limiting. For example, most engineered products function until components degrade or break, when they can be discarded or repaired. In biological turnover, by contrast, degradation frequently removes functional components. Similarly, intuition regarding energetic costs could be misleading when the underlying basis for a process such as turnover in a given context is unclear. Much attention has been drawn to unraveling the plentiful instances of regulation in biology, where mechanisms control response to stimuli. Such attention could yield the impression that regulatory surveillance mechanisms dominate biological systems, and this could take focus from the role(s) of turnover

in the first place. Some components in biology turn over in days and others perdure across decades—such as brain neurons, kidney cells, or cardiomyocytes, a difference that could also obscure the basis for turnover. Why does turnover vary so greatly across systems? For some processes, turnover has been suggested to prevent gaps and to replace degrading components.¹ In a number of contexts, however, turnover has been perceived to be bewilderingly rapid and its purpose treated as perplexing and seemingly futile, mainly for transitions, or varied case-specific possibilities have been proposed.

Life exists in harsh and variable conditions. Mechanical disruption, predation, and pathogens can inevitably eliminate system components, given enough time. Components can also accrue damage, without being lost, through wear and tear, chemical modification, or conformational aberration. Intrinsic variation and errors are also inevitable during the building and maintenance of systems. These facts present challenges to the viability of life. Short of replicating fast enough to outpace such disruption, life demands mechanisms to handle error, variation, and damage.

In this Perspective, I first outline a turnover theory in three parts: (1) turnover enables replacement of missing components, stability in levels despite variation and error, sporadic damaged component removal, minimization of chronic wear and damage accumulation, prevention of stable hazard formation, building and regulation of component numbers, and transitions; (2) recovery from damage and error can be accomplished by detection and repair (surveillance) or turnover (Figure 1A), and turnover can often be superior to exclusively surveillance-based alternatives; and (3) the presence and rate of turnover across systems is established by trade-offs between metabolic costs and



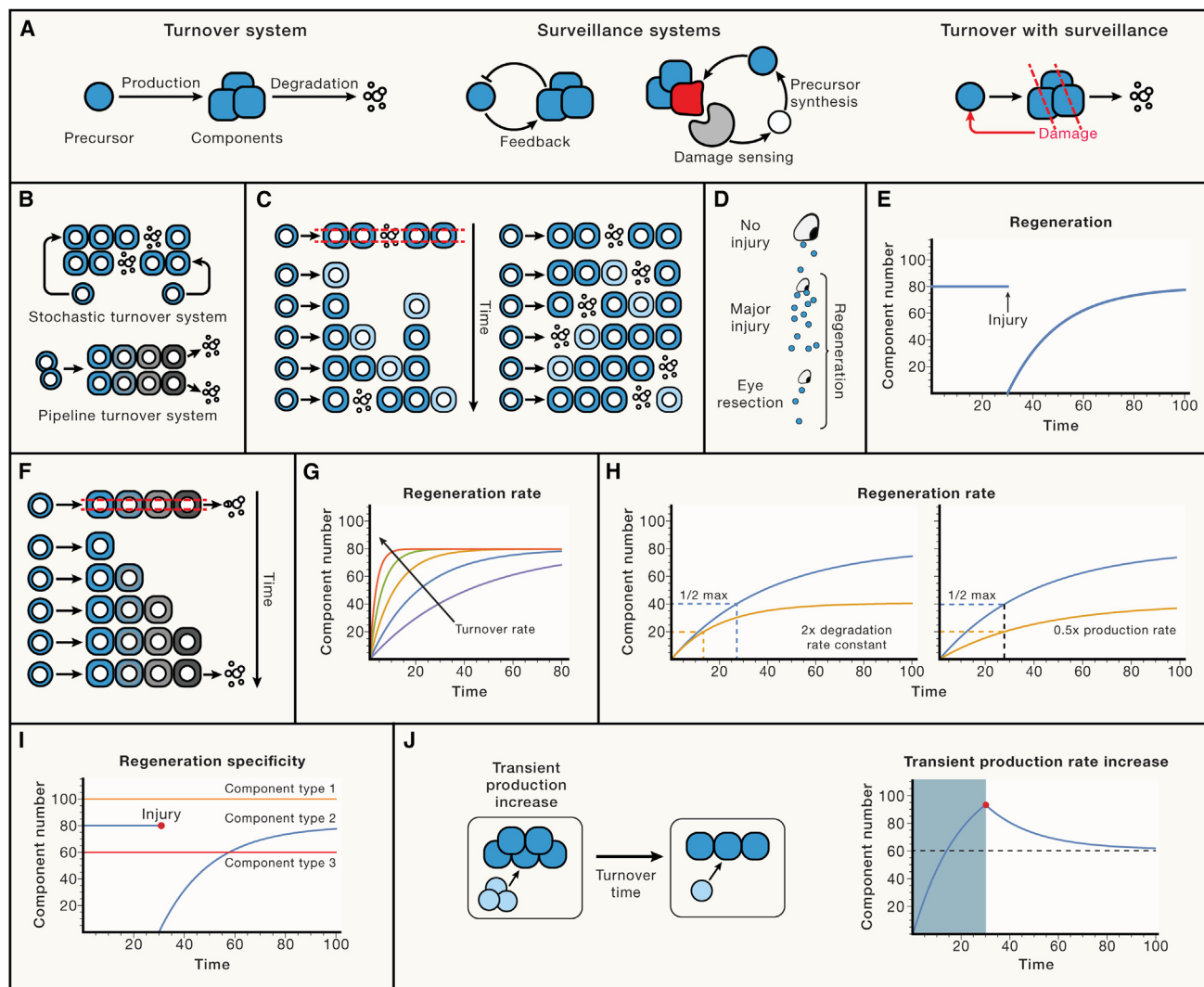


Figure 1. Turnover enables regeneration

(A) Two broad damage repair approaches, turnover, with continual production and degradation, and surveillance, such as with feedback inhibition on production or damage detection and targeted synthesis. Surveillance can also be added to a turnover system (right). Small white circles: degradation.

(B) Top, a stochastic turnover system using cells as components. Continual production and stochastic, age-independent degradation is depicted. Bottom, a pipeline turnover system with continual production and age-dependent component degradation.

(C) Turnover can enable regeneration, without necessitating damage detection, as a result of constant production and degradation of the system. Light blue cells represent those newly added over a given time interval.

(D) Example of regeneration through turnover: planarian eye resection does not increase eye progenitor production, but eyes regenerate nonetheless. Major injury (e.g., decapitation) amplifies proliferation generically.

(E) Regeneration after damage with a stochastic degradation system. Components with abundance N are produced with rate β and degraded with degradation rate $k_{\text{deg}}N$. (See text for details.) β , 4 and k_{deg} , 0.05.

(F) A pipeline turnover system also can regenerate after component loss without necessitating change of component production rate or lifespan.

(G) The production rate β , 2, and the degradation rate constant k_{deg} , 0.025, of different systems are scaled by a turnover factor b , 1, 2, 4, 8, 16; as b increases, regeneration occurs more rapidly.

(H) Left, two systems have the same production rate ($\beta = 2$), but the orange system has twice the degradation rate constant ($k_{\text{deg}} = 0.05$) as does the blue system. Right, two systems with the same degradation rate constant ($k_{\text{deg}} = 0.025$); the orange system has half the production rate ($\beta = 1$) of the blue system.

(I) Regeneration specificity can simply be explained with turnover without necessitating detection of which component type is missing ($k_{\text{deg}} = 0.05$, β , 5, orange; 4, blue; 3, red).

(J) Overproduction in development or following a transient production stimulus (e.g., damage); turnover can restore normal steady-state levels without necessitating a mechanism to detect precisely when production is complete. Right, $\beta = 6$ for $x \leq 30$ and $\beta = 3$ for $x > 30$; $k_{\text{deg}} = 0.05$.

system benefits, combined with constraints. I then describe diverse turnover contexts, supporting the argument that turnover is a pervasive solution to challenges of biology and seeking to

unite disparate, often unclear findings with candidate common underlying principles. Finally, I suggest that the limits of turnover form a basis for aging.

THE PROPERTIES AND BASIS FOR TURNOVER

Important concepts relevant for turnover date back decades. Prior to extensive data on turnover, biology was suggested to operate as an open system involving steady states.^{2–4} Open systems incorporate external energy to address the challenge of entropy, with life avoiding thermodynamic equilibrium.⁵ Theoretical work showed key properties of steady states, including that they can be reached from different starting points and can be returned to following stimulation or disturbance.^{2,3} In experimental evidence thereafter, Rudolf Schoenheimer used radio-labeling to recognize molecule turnover in what he referred to as a “dynamic state of body constituents”.⁶ Labeling studies, such as from Charles LeBlond,¹ recognized extensive turnover in certain cell populations. This pioneering work presaged decades of study on turnover mechanisms, including on gene expression, proteolysis, autophagy, cell division, apoptosis, and stem cells. In the following sections, how turnover can address distinct biological challenges is delineated. I suggest that defining these contexts, which could differ in their impacts across systems, is central to explaining both the purpose of turnover and the basis for its rate for distinct systems.

Properties yielded by turnover

Turnover can result in regeneration

Many systems display regeneration, or the capacity for missing component replacement. This definition can apply across scales to injuries removing cellular components, cells, or cell assemblages. A surveillance system that can detect component absence and order replacements for regeneration seems logical—how else could it work (Figure 1A)? For example, if a car part breaks, the broken component is detected, a replacement is ordered, and new part installation solves the problem. In a continually turning over system, by contrast, regeneration can occur without needing to detect that anything is missing at all,⁷ consistent with the classical concept of steady-state properties. One turnover system class involves constant production and stochastic component degradation (Figure 1B). Consider two such systems (X and Y), where one new component is added per minute and every component has a one-in-four chance of degrading per minute (Figure 1C). Both systems have on average four components at steady state (in every minute on average, one component is added and one randomly degrades). Injury removes all four components of system X. One minute later, a new component has arrived to system X. Over the second minute, the first component has only a one in four chance of degrading, and a second component arrives. After system X has added its fourth component, the system has been restored and the probability of a death event will equal the rate of addition over subsequent minutes. Systems X and Y both had the same production rate, yet one did not change in size and the other regenerated.

Planarian eye regeneration presents an example of regeneration through turnover. Planarians are flatworms that regenerate with pluripotent stem cells called neoblasts.⁸ The entire planarian body is in a continuous state of turnover with neoblasts constitutively producing progenitors for all tissues, from neurons to intestine to skin.^{8,9} Planarian eyes, for example, are completely turned over in a month or two.^{7,10} A transplanted

neoblast, through tissue turnover and regeneration, can convert the entire host genotype to that of the donor cell.¹¹ Head amputation triggers a burst in eye progenitor number, associated with rapid eye regeneration.¹² This could lead to intuition that the need to regenerate eyes is sensed and a suitable response is mounted. Following eye-specific resection, however, no change in eye progenitor production occurs, yet eyes regenerate nonetheless.⁷ This can be explained by the constitutive eye cell turnover that occurs even without injury. Following single-eye resection, both eyes receive progenitors at this low constant rate, yet one grows (regenerates) and the other stays a constant size⁷ (Figure 1D). Large injuries could generically increase stem cell proliferation to simply accelerate regeneration, being added to turnover.⁷

In a stochastic degradation turnover system, production of N components with rate β and stochastic component degradation with rate $k_{\text{deg}}N$ yields an equation for component dynamics.

$$\frac{dN}{dt} = \beta - k_{\text{deg}}N$$

Degradation proceeds as a first-order reaction, with rate dependent on N because each component has some probability of degradation per unit time, expressed as a degradation rate constant k_{deg} , regardless of component age. In such a system, perfectly good components are destroyed, including some just synthesized. Such an equation is an example of a steady-state scenario and has often been applied to protein levels.^{13,14} Solving this equation gives

$$N_t = \frac{\beta}{k_{\text{deg}}} (1 - e^{-k_{\text{deg}}t}) + N_0 e^{-k_{\text{deg}}t}$$

After removal of components, this system regenerates, with the rate of accumulation decreasing as the steady state is approached (Figure 1E). The steady-state value of N is

$$N_{\text{steady state}} = \frac{\beta}{k_{\text{deg}}}$$

These equations result from a continuous approximation of component production and destruction, which can serve reasonably well if component number is large. Synthesis and destruction events can alternatively be considered a Markov process, where the steady-state value is expected to show a Poisson distribution.¹⁵ When system component number is small (as in Figure 1C), component number fluctuation around a steady state is therefore predicted.

In stochastic turnover systems, components degrade whether they are young or old, damaged or not. Examples include most cellular proteins, bacterial cell walls, and ribosomes (see the section [The ubiquity of turnover](#)). Young and old proteins have similar probabilities of degradation, following first-order kinetics and a labeled cohort display exponential decay.^{13,14} In a global analysis, ~14% of proteins displayed non-exponential decay¹⁶; however, in these cases there was a higher, rather than lower, probability of decay in young proteins (during synthesis or early thereafter), followed by a constant probability of decay over age.¹⁶ Proteins, in general, therefore do not appear to be degraded because they are getting old.

Other turnover systems show average component lifespan (component loss probability increasing with age) as opposed to stochastic degradation (Figure 1B). Such a system could be considered a pipeline turnover system. Pipeline examples include terrestrial skin and shark teeth; mRNA decay probability also increases with time associated with the degradation mechanism (see the section [The ubiquity of turnover](#)). Pipeline turnover systems also regenerate. After component loss, new young components are added at a constant rate and do not initially degrade because they are young. As time progresses, initial components become older and eventually degrade; at this point the system has regenerated without necessitating surveillance (Figure 1F). With component lifespan L , steady-state component levels will be

$$N_{\text{steady state}} = \beta L$$

Such a system can be established even when no damage is accrued in most components over their lifetime. In other words, component removal in many pipeline systems is programmatic rather than damage based.

If regeneration relies on turnover, it will occur faster as turnover becomes more rapid. For example, if production and degradation rates increase by the same factor for a stochastic system, the steady-state component amount remains constant, but the regeneration rate increases (Figure 1G). A known attribute is that the time to reach steady state is heavily dependent on the degradation rate (Figure 1H). Various types of regulation are frequently layered on top of systems that utilize turnover for regeneration. For example, tissue injury can trigger generic upregulation of cell proliferation.⁸ Notably, regeneration-specific processes not associated with turnover also exist, such as dedifferentiation¹⁷ or feedback from target cells.¹⁸ For a system relying on turnover for regenerative repair, however, a high price of rapid turnover might need to be paid.

Regeneration specificity

For a system with multiple component types (cells or molecules), a major challenge is determining which component is missing and regenerating it without increasing production of other uninjured components—regeneration specificity.⁷ Turnover presents a simple solution to this substantial challenge: the missing component type will specifically grow without requiring change in the production or degradation rates of any component (Figure 1I).

Determining when regeneration and development are complete

How do systems specify when regeneration or development is complete? One option is to utilize a production pulse; however, for regeneration, tuning pulse magnitude to the damage amount could be prohibitively challenging. A surveillance system would need to precisely measure the damage and to act on a potentially noisy production system (e.g., transcription or cell production). Alternatively, production could be induced until no damage is detected, quenching a hypothetical signal. However, signal cessation could leave a pipeline of precursors en route and consequent overshooting (Figure 1J). It might seem paradoxical to start destroying components before building is complete, but turnover is one solution to reaching a desired abundance. For

example, flagella regeneration rate slows as the final size is neared,¹⁹ and cell death events occur in regenerating planarian eyes.⁷ If regulation is added to turnover, such as a transient damage-induced production increase, any incidental overshooting will be sculpted away by turnover (Figure 1J). Similarly, developmental programs could in principle achieve a final size and address overshooting or undershooting once the system enters a turnover regime.

Turnover enables state maintenance (intrinsic variation and error)

Intrinsic fluctuations and errors are fundamental challenges to biology. Extensive turnover can yield a steady state and address these challenges, providing a generic path to return from deviation (Figure 2A). This can also help prevent error or variation, such as with regulatory factors, causing compounding effects. For example, stochastic variations in gene expression levels occur, such as with transcriptional bursting²⁰ (Figure 2B). Turnover can result in fluctuation around mean mRNA and protein values; faster turnover of mRNA than protein can minimize protein level correlation with fluctuating mRNA.²⁰ Stochastic variation can also result from reactions with molecules at low abundance.²¹ Random or semi-random assortment of components during cell division can result in partitioning errors, with daughter cells inheriting different concentrations of factors.²² Continuous production and degradation would return cells to the mean (Figure 2B). Bacterial cell size can vary during division and presents an example where constant addition and loss, here through dividing in half, can passively return size to the mean, rather than necessitating an active corrective process²³ (Figure 2B). The environment in which processes occur is also variable. Reactions or interactions in or between cells can have error rates or off-target products. Such products (along with functional products) become gradually cleared, non-selectively, through random turnover. Additional properties, such as negative feedback, can be added to buffer against variation and add robustness.^{20,24}

Removing errors and variation during building

Errors or variation inevitably occur when building an entity. For example, genes might be transcribed in the incorrect cell type or position during development or regeneration. Similarly, cell identity choices might occur in the wrong ratios or locations. In principle, complex regulatory machinery could be evolved specific to developmental processes to recognize and correct such errors. Because of their unpredictable nature, however, it could be a challenge to evolve detection systems for all possible scenarios. By contrast, once the system enters turnover, prior errors could readily be fixed without needing regulation (Figure 2C). Spatial transcriptomics opens the door to detailed study of these possibilities.

Maintaining a low level of damage (chronic wear and damage)

Systems can have predictable basal rates of damage occurrence through continual wear, tear, and damaging agent production/exposure. For example, macromolecular complexes might slowly accumulate oxidative damage, or bird feathers might continually acquire mechanical damage, gradually diminishing function. Surveillance systems limited in ability to detect weak impairment would result in a continual state of disrepair.

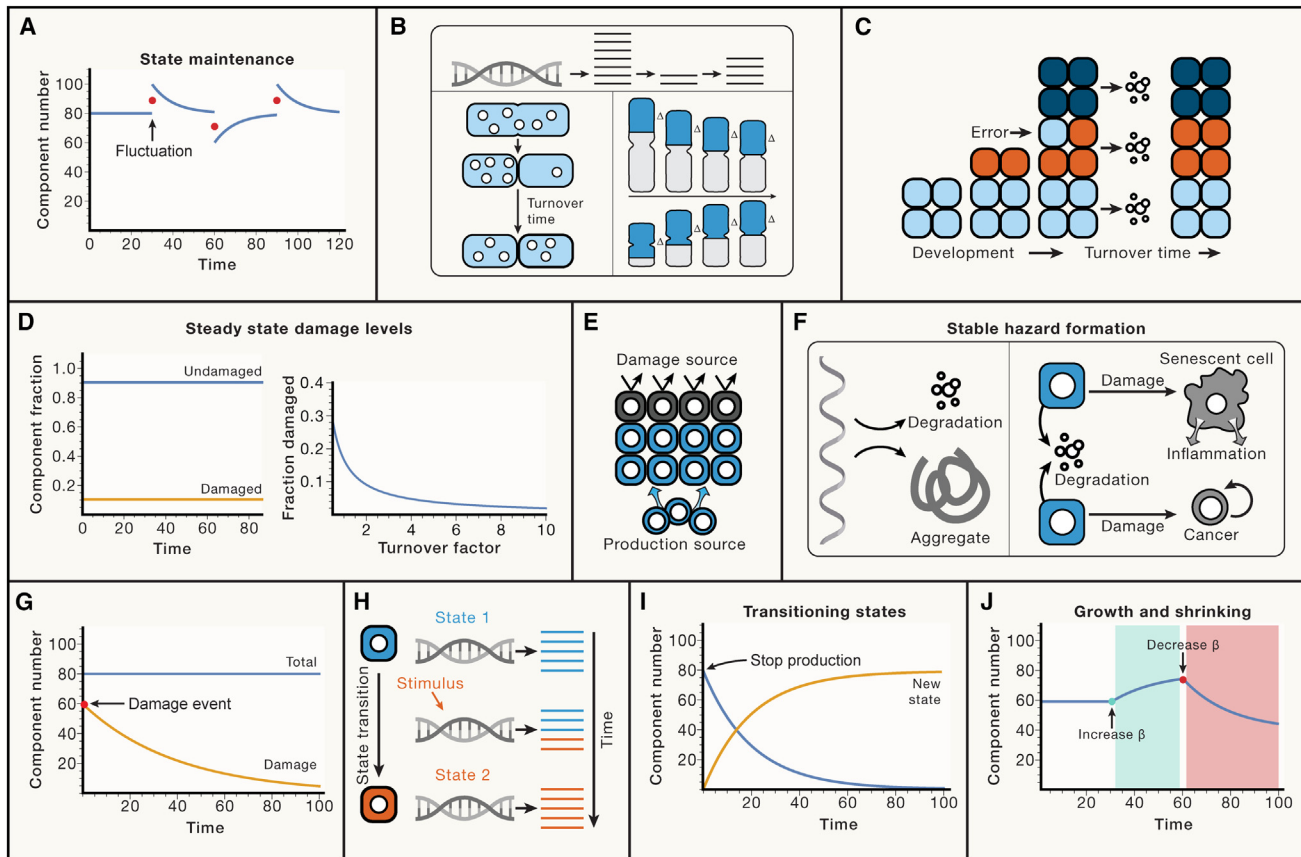


Figure 2. Turnover enables state maintenance without damage accumulation and transitions

(A) Fluctuations and errors are inevitable. Turnover can return systems to the steady state without necessitating detection systems. $\beta = 8$ and $k_{deg} = 0.1$; fluctuations of 30 components were introduced every 30 time units.

(B) Examples of variation in levels; top, transcriptional bursting with different transcript numbers over time; bottom left, partitioning errors of a component (dots) during cell division return to the mean with turnover; bottom right, bacterial cell size variation can return to the mean with adding a constant amount per cycle (blue region marked with a delta) and dividing evenly.

(C) Errors in development, such as regional cell-type identity (depicted as different colors), relative cell-type abundance, or errors in gene transcription in a particular location, are readily pruned with turnover.

(D) Left, with constitutive damage, the fraction of damaged components in a turnover system can be kept low rather than accumulate. N can convert to a damaged version N^* with rate $k_{dam}N$, and N and N^* entities degrade with the same rate constant k_{deg} . Steady-state values for N and N^* are $N_{ss} = \frac{\beta}{k_{deg} + k_{dam}}$ and $N^*_{ss} = \frac{\beta k_{dam}}{k_{deg} + k_{dam}}$. Values used: $\beta = 4$, $k_{dam} = 0.005$, $k_{deg} = 0.05$. Right, the steady-state damaged fraction value depends on the turnover rate of the system, observed by scaling the production rate and the degradation rate constant with a turnover factor b from 0.5 to 10, with $\beta = 4$, $k_{dam} = 0.01$, and $k_{deg} = 0.05$ values shown.

(E) In a system where components (e.g., a barrier structure) are continually exposed to damaging conditions, sequestering component production (e.g., stem cells) to a comparably protected region can help replace components with undamaged ones.

(F) Stable hazards, such as protein aggregates (left) or senescent or cancerous cells (right) can form following sufficient component damage accumulation, and be difficult to degrade.

(G) When a fraction of system components becomes sporadically damaged (e.g., transient chemical exposure), the damaged fraction is readily cleared with turnover and replaced with undamaged components without surveillance needed. Blue, total components, damaged (N^*) and undamaged (N), stay constant with $\beta = 2$ and $k_{deg} = 0.025$. The damaged components, with $N_0 = 60$ and with $k_{deg} = 0.025$, steadily decline.

(H) Transitioning between two states is facilitated by turnover, where unwanted components (blue mRNA species are shown) are degraded without necessitating specific targeting for removal.

(I) State transition from blue to orange components; blue $N_0 = 80$, $\beta = 0$, $k_{deg} = 0.05$; orange $N_0 = 0$, $\beta = 4$, and $k_{deg} = 0.05$.

(J) Increasing or decreasing production can tune system size to varying conditions or in evolution; $N_0 = 60$, $k_{deg} = 0.05$, and $\beta = 3$ for $t \leq 30$, 4 for $30 < t \leq 60$, and 2 for $t > 60$.

By contrast, with turnover set to a suitably high burn rate, a system can be kept in a youthful, fit state without needing to know anything about how components are faring (Figure 2D). The turnover rate will strongly impact the steady-state fraction of damaged system components (Figure 2D). Systems exposed to continually damaging conditions (e.g., barrier struc-

tures) might thus be forced in evolution into a shockingly high turnover rate. The mammalian intestine presents a dramatic case, turning over as rapidly as in two to four days.^{25–29} This rate is staggering considering other cells (e.g., neurons) can live for decades. Sequestering a component source (e.g., stem cells) from damaging conditions (Figure 2E) can enable

mature components exposed to chronic damage to become replaceable.

Lowering risk of stable hazard formation

In some systems, accumulated damage could create specific challenges if certain aberrant components or states could arise that are stable and hazardous (Figure 2F). For instance, misfolded proteins or RNA could trap additional factors, resulting in aggregates that could be hard to degrade and become cytotoxic.³⁰ Similarly, cells can become senescent, secrete damaging factors, and become hard to remove,³¹ or they could become cancerous. Systems with such stable hazard risk must either evolve mechanisms to recognize and eliminate hazards or must minimize forming them in the first place, which can be accomplished with rapid turnover.

Eliminating damaged components (sporadic damage)

Cells can encounter sporadic, unpredicted damaging conditions that functionally impair but do not eliminate components. Repair here requires some system that does not rely on detecting component absence or some particular form of injury (not regeneration). For example, there are more than 35 ways for proteins to be oxidized,³² and lipids can incur oxidative damage or hydrolysis, lipid degradation products can modify proteins or nucleic acids,³³ infections could inflict novel damage, or mutagenic conditions could be encountered. Numerous particular damage-detection systems exist and often interface with normal turnover machinery; however, random turnover eventually clears damaged components even if the nature or degree of the damage is undetectable or novel or if there has been insufficient opportunity to evolve a collection of response systems for all cases (Figure 2G).

Enabling transitions and regulation

In the context of protein turnover, a commonly hypothesized purpose for rapid turnover is to facilitate regulation.¹⁴ For example, Alon (2020) states that “[t]he benefit of such a futile cycle is fast response times once a change is needed.”³⁴ Rapid and ongoing turnover even in non-dividing bacteria was proposed to allow for rapid enzyme production in changing conditions, for instance,³⁵ or more generally to facilitate regulation.^{36–39} As a cell transitions in state (e.g., differentiation or under stress), a portion of its transcriptome and proteome needs to transition, with turnover making the system agile. Regulation can initiate state transition and turnover can enable it by removing components without necessitating complex programs to recognize and remove unwanted components (Figures 2H and 2I). For example, changes in protein levels during *Xenopus* development and embryonic stem cell differentiation can be simply modeled with constant degradation rates; adding microRNA-mediated action can improve the model for some genes.^{40,41} Turnover can also enable dynamic system-size regulation. If production or degradation rates are shifted by cues, such as growth signals, then system size can readily adjust (Figure 2J). Despite turnover mediating transitions, it is clear that many components not involved in dynamic changes turn over nonetheless.

Other considerations and properties

Establishing and evolving relative proportions. Tuning production/degradation rates presents one path to evolving component (molecules/cells) proportions. Alternative solutions exist, such as differential growth.

A mechanism for gradient formation. Signaling protein (morphogen) gradients can instruct regional cell-fate adoption. Constant ligand production and diffusion can be coupled with degradation to realize such a gradient.

New modalities in evolution can rapidly experience the benefits of regeneration and stability. Placing some new system component into an existing generic turnover process (e.g., a novel cell type or new protein) immediately yields regeneration and a state maintained with high-quality components. Regulatory bells and whistles can evolve over time.

Turnover is compatible with the emergence of myriad additional regulatory features. Biology involves lots of regulation, which can be layered onto, or utilize, turnover by changing production or degradation. Regulation of continuous degradation has also been harnessed for certain responses. For example, Wnt signaling involves inhibiting continuous β -catenin degradation.⁴²

Properties yielded by turnover: Advantages over surveillance

How do we reconcile the benefits of turnover with the high costs of continually degrading functional components—why not just identify and replace dysfunctional or missing components?

Repair sensitivity and specificity

Determining that a component is impaired, then mounting a specific response with suitable magnitude is a substantial challenge. For precision, surveillance needs to be sophisticated enough to recognize that damage is present, even if subtle, a mechanism to signal for production of the specific damaged component and the right amount of it, and to have a mechanism to incorporate the replacement component where needed (Figure 3A). For example, car repairs can involve computer-based diagnostic tests for problem identification, a replacement part ordering mechanism, and an expert mechanic to replace the dysfunctional component. Such sensitivity and specificity might often be more readily achieved in evolution through turnover.

A large number of systems and component types to maintain

The challenges of sensitivity and specificity attend the repair needs for any one component, and some surveillance systems have evolved. However, if biology primarily relied on surveillance for regulating component repair and levels, an enormous array of surveillance systems capable of detecting each potentially damaged entity and various damage class types might be needed. This scale problem alone could preclude surveillance being the primary solution to the challenges of perturbation, error, and variation in biology.

As an example of the challenges of specificity, sensitivity, and scale of component types for repair, a variety of studies indicate a general lack of surveillance that feeds back on specific protein production. In bacteria, yeast, and other eukaryotes, protein production rates are generally proportional to their stoichiometry in complexes (proportional synthesis).^{43,44} Elimination of one copy of 730 genes in diploid yeast indicated that only <5% of genes showed compensatory changes with protein levels above 50% of the wild type.⁴⁵ Aneuploidy experiments and wild copy-number variants showed that protein levels typically scale with gene copy number.^{46,47} A minority of proteins show some

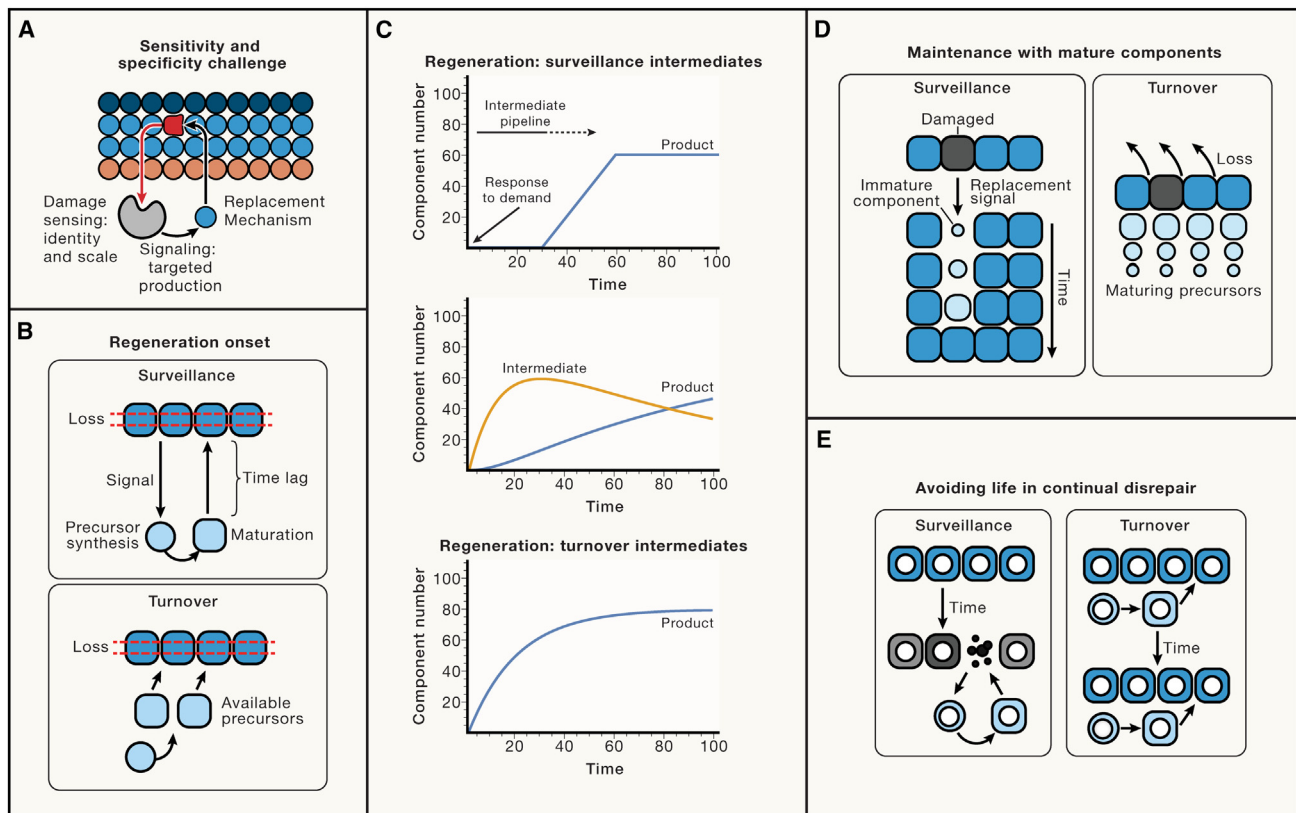


Figure 3. Challenges of surveillance met by turnover

(A) A suggested challenge for exclusively surveillance-based repair is sensitivity and specificity in detecting which component is damaged, especially when damage is small or system dysfunction minimal. A mechanism for error detection (red arrow to gray sensor) signaling precursor formation (black arrow), and replacing the damaged component (black arrow) could be needed.

(B) Surveillance scenarios where injury (red dashed line) removes components and triggers synthesis of a precursor that must mature or transit to replace the missing component, creating a repair time lag.

(C) Top, precursor synthesis initiates after injury and lasts for 30 time units; precursors undergo a transit time of 30 units before maturing to a stable product. Middle, injury stimulates production of a precursor P from some activated source component A as $k_1 A$ and P is converted to stable product N as $k_2 P$, such that $N(t) = A_0 \left[1 - \frac{k_2}{k_2 - k_1} e^{-k_1 t} + \frac{k_1}{k_2 - k_1} e^{-k_2 t} \right]$, $k_1 = 0.08$ and $k_2 = 0.01$ depicted. Bottom, regeneration in a turnover scenario where a precursor P is synthesized with rate k_1 , P is converted into product N with rate $k_2 P$, and N is degraded with rate $k_{deg} N$. The condition where the system is at steady state and injury removes a portion of N but does not affect the precursors is considered, with P_0 equal to the steady-state value of P such that thereafter $N(t) = \frac{k_1}{k_{deg}} (1 - e^{-k_{deg} t}) + N_0 e^{-k_{deg} t}$, with k_2 not impacting $N(t)$ and values $k_1 = 4$ and $k_{deg} = 0.05$ shown. In this case, repair can initiate with available precursors without delay.

(D) Left, surveillance system where damage or loss triggers precursor synthesis; the system endures with an immature precursor over time. Right, turnover system with precursor maturation (shown as pipeline); renewal with mature components.

(E) Left, a hypothetical repair system (here, cells) where only fully dysfunctional or absent components are detected. Other components will deteriorate and the system will continually have poor quality components. Right; a continually turning over system can be kept in a perpetually fit state.

compensation; these are enriched for components of complexes and changes mainly occurred to degradation rather than production.^{44,46} Similarly, increased copy number in yeast resulted in compensatory changes in protein degradation for only a fraction ($\sim 10\%$) of tested genes and not in synthesis rates.⁴⁸ For some cases where posttranslational compensation occurred, complex subunits could have faster degradation of free components.^{16,49} For instance, overproduction of particular ribosome proteins leads to more rapid degradation of free protein than of ribosome-associated protein.⁵⁰ These findings suggest that when protein levels are altered, generally speaking, cells do not know anything is wrong. Turnover of the proteome, however, can handle problems without the system needing to know a problem is there.

Rapid regeneration onset and system maintenance with mature components

With turnover, components are constantly being made, such that replacements can be at hand when damage strikes. This can enable rapid regeneration onset compared to the alternative of *de novo* synthesis, for systems where component production or maturation (e.g., progenitor cells) takes substantial time (Figures 3B and 3C). Blocking new synthesis would block regeneration initiation in a system relying on regulation (such as feedback), but repair initiation could proceed with intermediate progenitors en route anyway in a turnover system. For example, blocking protein production does not stop initial flagella regeneration in *Chlamydomonas*.¹⁹ Waiting for new components after surveillance could yield

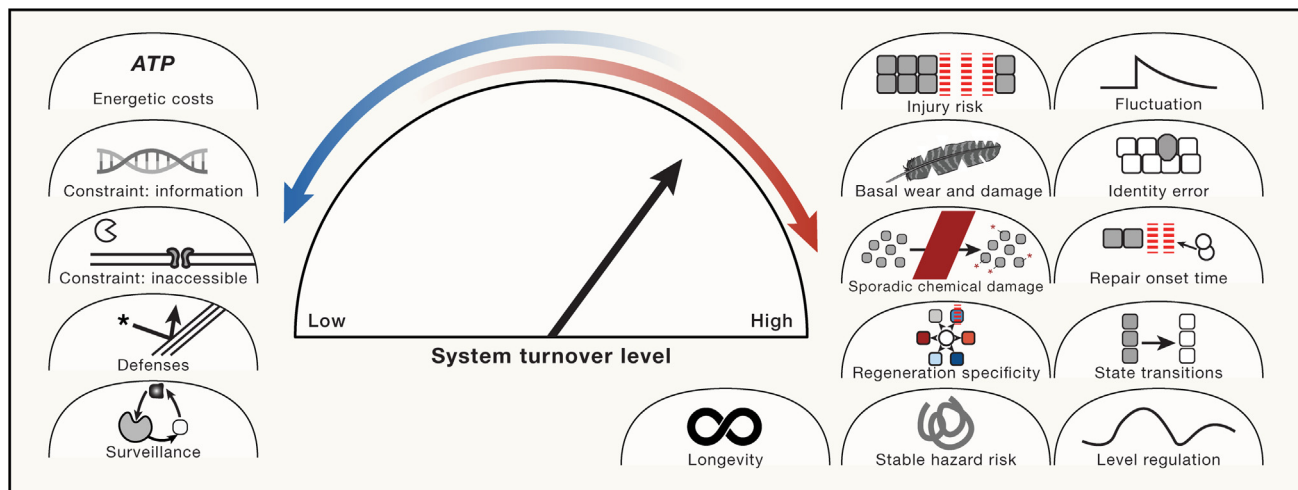


Figure 4. Trade-offs and constraints can set the presence and rate of turnover across systems

The turnover rate of components within different, modular biological systems could be tuned in evolution by trade-offs and constraints. On the negative side, the metabolic costs of component production and/or degradation pushes turnover rates lower. Constraints on the compatibility of turnover with system functioning or accessibility of components to turnover can also restrict or push toward lower turnover. Successful surveillance and defenses can also lower investment. On the positive side, a series of benefits, impacted by fitness impacts and rate parameters, pushes a system toward higher turnover levels. See sections on factors restricting or promoting turnover presence and rate for delineation of factors depicted.

catastrophic outcomes for certain systems (imagine driving a car without a steering tie rod) and thus be inviable as an alternative to turnover.

For a system that faces constant deterioration, component maturation time (e.g., for some cell types) presents a similar challenge for surveillance. Once surveillance has detected the need for replacement and signaled for the new component, immature replacement components might be part of the system (Figure 3D). For example, shark teeth have a pipeline of maturing replacements ready if a tooth is lost, or for replacing teeth that are simply worn or damaged.

Together, the limits on problem detection sensitivity and specificity, potential time lags in replacement component production, and immature components could leave a surveillance-dominant system in a state of continual disrepair and make this alternative simply inviable (Figure 3E). For example, if repair required some accumulated damage threshold (or system failure) to be detected, then such a system could constitutively be populated with poorly functioning components.

Surveillance systems can have limitations and themselves be systems to maintain

Surveillance-like systems, such as negative feedback, can in principle dampen noise but also present their own challenges; for instance, they could potentially amplify variation depending on reaction details.⁵¹ Many surveillance systems that can be envisioned involve particular molecular machinery or cell types that themselves might necessitate repair or surveillance processes to be maintained, creating an ever-expanding need for surveillance systems. This complexity could limit surveillance systems from being a primary means for maintaining components. By contrast, the destruction machineries of turnover, such as proteasomes,⁵² can themselves readily be maintained by turnover, solving this dilemma.

The presence and rate of turnover could be established by trade-offs between metabolic costs and system benefits and constraints

Some biological materials can be quite stable without processes to degrade them, such as bone, hair, or peptide bonds.⁵³ It takes energy to maintain steady states,^{3,5,54} and alternatives to turnover can yield certain desired properties.¹⁸ Furthermore, turnover rates vary widely, needing explanation. Key contextual features thus likely influence turnover prominence across systems. I suggest that trade-offs involving multiple parameters and constraints (enumerated in the following sections) underlie the presence and level of turnover across systems, with evolution maximizing lifetime organism reproductive output based on these parameters (Figure 4).

Factors restricting turnover presence and rate

Despite turnover upsides, some components are not replaced. What forces push against turnover? (1) The most obvious downside of turnover is its energetic cost. (2) Turnover of a particular component (or certain turnover rates) could be incompatible with system function, representing a constraint. For example, neuron turnover could have negative impacts on experience-modulated properties. Turnover of DNA or of the nucleus in post-mitotic cells could be incompatible with fidelity of information retention. (3) Some components could be less accessible to evolved turnover machinery, providing a different constraint type. For example, lens crystallin,⁵⁵ the nuclear pore complex,⁵⁶ and collagen⁵⁷ are all long-lived proteins. Degradation (e.g., of a buried/isolated complex) or synthesis (e.g., incompatibility of a tissue with stem-cell maintenance) might limit turnover for such cases. (4) Investments that lower damage and error probability (e.g., defense; mucous, antioxidative processes) can lower turnover rate, if achievable and of low enough cost. (5) Surveillance, where achievable, can offer substantial energy savings. If damage probability is low and surveillance can be evolved,

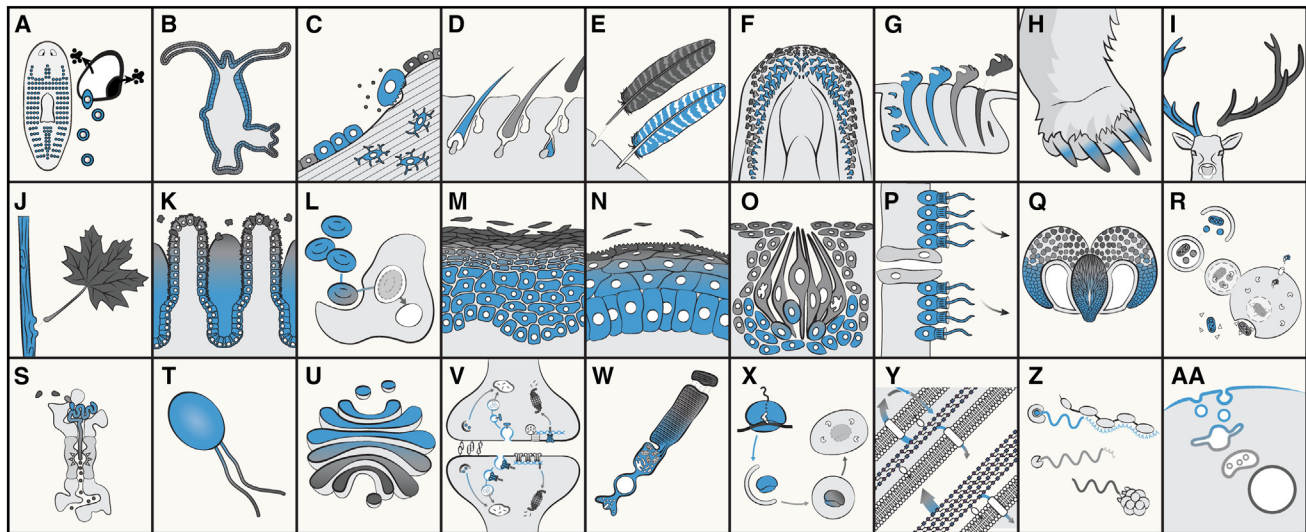


Figure 5. Turnover systems

Blue shading indicates production areas, and gray indicates maturation, recycling, or degradation. (A) Planarians and (B) hydra turnover their entire bodies; (C) bone; (D) hair; (E) feather; (F) catfish teeth, adapted from Fraser et al. (2020)⁵⁸; (G) annelid chaetae, adapted from Hausen (2005)⁵⁹; (H) American badger claws; (I) antlers; (J) leaf abscission; (K) small intestine; (L) RBC degradation; (M) skin basal keratinocytes produce the outer layer (stratum corneum) where cells are shed (desquamation); (N) cornea basal progenitors, wing cells, and superficial squamous cells that slough off; (O) taste bud bipotent basal progenitors produce taste cells and surrounding keratinocytes; (P) sponge choanocytes; (Q) ctenophore tentacle root: colloblast stem cells, distal-lateral and muscle progenitors, central base; (R) autophagy; adapted from Kaur and Debnath (2015)⁶⁰; (S) ubiquitin-proteasome system, adapted from Prakash and Matouschek (2004)⁶¹; (T) *Chlamydomonas* flagella; (U) Golgi; (V) pre-synaptic and post-synaptic protein turnover; autophagy and the ubiquitin-proteasome system are depicted; (W) photoceptor rod; membrane discs are continually assembled at the base, progress, and are shed; (X) ribosome degradation through autophagy; (Y) peptidoglycan (PG) cell wall turnover; gram-negative (upper left) and gram-positive (bottom right) bacteria. Squares, GlcNAc. Hexagons, MurNAc. Ovals, lipoprotein. Adapted from.⁶² PG fragments are either lost or recycled; (Z) mRNA: 5' cap bound by eIF-4E and poly-A tail bound by Poly-A binding protein (PAB). CCR4-NOT, deadenylation; 5' cap removal, Dcp1-Dcp2 complex; degradation with 5'-3' XRN1 exonuclease or 3'-5' RNA exosome exonuclease. Adapted from⁶³; (AA) plasma membrane turnover by endocytosis; membrane and contents can fuse with the lysosome.

damage-associated surveillance can dominate (e.g., de-differentiation in certain regeneration contexts). Surveillance can be specific or generic; generic surveillance changes production or degradation broadly without specificity to damaged component identity or levels. Intermediate scenarios are common and involve turnover plus regulation, such as with multiple cell stress-response systems, generic proliferation increase after major planarian injury, or degradation of damaged mitochondria.

Factors favoring turnover presence and rate

Parameters related to the fitness impact and rate of damage, error, and variation can determine system turnover rate and can vary across systems: (1) the frequency of injuries that remove components, necessitating regeneration; (2) the rate of constitutive component damage occurrence and wear and (3) system fitness impacts for different levels of such damage; (4) the rates and (5) system fitness costs for classes of sporadic or unpredictable (e.g., chemical) damage, not eliminating components (not a regeneration scenario)—this can be impacted by evolved behaviors lowering risk; (6) the importance of specificity in repair (regeneration or chemical damage); (7) the risk that stable hazard classes impose and (8) their formation rates, if this can occur when damage is allowed to accrue; (9) system fitness sensitivity to component level fluctuation during state maintenance and (10) to inaccurate component levels or identities when building, such as in regeneration or development; (11) system tolerance to time-lag length for component

replacement after damage or loss; and (12) sensitivity to immature component presence during repair. Other parameters relate to needs for dynamic change: (13) the probability of state transitions and the importance of transition timescale; and (14) the frequency with which system size needs to adjust and the importance of response time. (15) Selection toward longevity for the life-history strategy of a given species could also push toward investment across systems. (16) Seasonal changes or need for increased component size during growth could also result in turnover. There could also be interactions between these parameters. For example, high turnover rates selected for one reason (e.g., high extrinsic damage) could yield pleiotropic benefits, such as regeneration and system longevity. Furthermore, degradation selected for one component (e.g., some pipeline cell type) might lead to turnover for other physically associated components. As discussed in the section [Turnover and aging](#) I suggest that the evolved turnover levels across systems can promote longevity and that turnover limits fundamentally contribute to the basis for aging.

THE UBIQUITY OF TURNOVER

In the following section, I describe programmatic turnover across scales, from large to small ([Figure 5](#); rates in [Table 1](#)). This section seeks to highlight underappreciated scope and rates of turnover traversing fields and the properties turnover can yield across contexts.

Table 1. Timescale of turnover across biological systems

component	organism, context	turnover time a, half-life; b, turnover; c, lifespan	Reference
mRNA	<i>Mus musculus</i> (fibroblasts)	2.1 h ^a	Eisen et al. ⁶⁴
proteins	<i>Mus musculus</i> (myoblasts)	2 days ^b	Cambridge et al. ⁶⁵
liver proteins	<i>Mus musculus</i> (liver)	3 days ^c	Price et al. ⁶⁶
proteins	<i>Saccharomyces cerevisiae</i>	10 h ^a	Claydon and Beynon and Helbig et al. ^{38,67}
cortex proteins	<i>Mus musculus</i> (cortex)	3–13 days ^c	Fornasiero et al. ⁶⁸
184 synapse proteins	<i>R. norvegicus</i> (hippocampus)	5 days ^a	Dörrbaum et al. ³⁹
miRNAs	<i>Mus musculus</i> (fibroblasts)	34 h ^a	Kingston and Bartel ⁶⁹
tRNAs	<i>Saccharomyces cerevisiae</i>	9 h to days ^a	Gudipati et al. and Phizicky and Hopper ^{70,71}
plasma membrane	<i>Mus musculus</i> (fibroblasts)	24–48 h ^a	Warren and Glick ⁷²
ribosomes	<i>Rattus norvegicus</i> (liver)	5 days ^a	Hirsch and Hiatt ⁷³
rod photoreceptor membranes	<i>Macaca mulata</i> - rhesus	9–12 days ^b	Young ⁷⁴
bacterial cell wall	<i>Bacillus subtilis</i>	50%/generation	Mauck et al. ⁷⁵
bacterial cell wall	<i>Escherichia coli</i>	50%/generation	Goodell ⁷⁶
titin	<i>Gallus gallus</i>	3–5 days ^a	Isaacs et al. ⁷⁷
mitochondria	<i>Rattus norvegicus</i> (liver)	10.3 days ^a	Fletcher and Sanadi ⁷⁸
flagella	<i>Chlamydomonas reinhardtii</i>	0.011 micron/sec	Marshall and Rosenbaum ⁷⁹
ER membrane	<i>Rattus norvegicus</i> (liver)	48 h ^a	Lee et al. ⁸⁰
small intestine	<i>Homo sapiens</i>	2–3.3 days ^b	Bertalanffy and Nagy and Lipkin et al. ^{28,81}
epidermis	<i>Homo sapiens</i>	39–45 days ^b	Weinstein et al. and Bergstresser and Taylor ^{82,83}
epidermis	<i>Danio rerio</i>	19.5 days ^b	Chen et al. ⁸⁴
red blood cells	<i>Homo sapiens</i>	120 days ^c	Shemin and Rittenberg ⁸⁵
corneal epithelium	<i>Homo sapiens</i>	7 days ^b	Hanna et al. ⁸⁶
taste bud receptors	<i>Rattus norvegicus</i>	10 days ^c	Beidler and Smallman ⁸⁷
tongue keratinocytes (papilla)	<i>Rattus norvegicus</i>	2–4 days ^b	Hume and Potten ⁸⁸
esophagus	<i>Homo sapiens</i>	9 days ^b	Bell et al. ⁸⁹
stomach	<i>Rattus norvegicus</i>	3.1 days ^b	Karam and Leblond ⁹⁰
colonic epithelium	<i>Homo sapiens</i>	3–4 days ^b	Lipkin et al. ⁸¹
rectal epithelium	<i>Homo sapiens</i>	6–8 days ^b	Cole and McKalen ⁹¹
trachea luminal secretory cells	<i>Mus musculus</i>	14 days ^c	Watson et al. ⁹²
trachea epithelial cells	<i>Mus musculus</i>	6 months ^a	Rawlins and Hogan ⁹³
sebaceous glands	<i>Homo sapiens</i>	7.4 days ^b	Epstein and Epstein ⁹⁴
osteoblast	<i>Mus musculus</i>	12 days ^c	Weinstein et al. ⁹⁵
regulatory T cells	<i>Homo sapiens</i>	12 days ^c	Borghans et al. ⁹⁶
dendritic cells	<i>Mus musculus</i>	5–7 days ^a	Liu et al. ⁹⁷
circulating neutrophils	<i>Homo sapiens</i>	19 h ^a	Lahoz-Beneytez et al. ⁹⁸
total B cells	<i>Homo sapiens</i>	52–76 days ^c	Borghans et al. and Macallan et al. ^{96,99}
olfactory bulb granule cells	<i>Mus musculus</i>	50%/year	Imayoshi et al. ¹⁰⁰
hair	<i>Homo sapiens</i>	10.8–38.1 months ^c	Courtois et al. and Saitoh et al. ^{101,102}
hair	<i>Mus musculus</i>	3–4 weeks ^c	Pennycuik and Raphael ¹⁰³
teeth	<i>Triakis semifasciata</i> - shark	9–12 days ^b	Reif et al. ¹⁰⁴
teeth	<i>Allosaurus</i>	100 days ^c	D'Emic et al. ¹⁰⁵
bone	<i>Homo sapiens</i>	10% skeleton/year	Frost ¹⁰⁶
antlers	cervidae	1 year ^b	Goss ¹⁰⁷
feather	bird species	molt 1–3 times/year	Lin et al. ¹⁰⁸
claw	<i>Taxidea taxus</i> - badger	130–140 days ^b	Ethier et al. ¹⁰⁹

(Continued on next page)

Table 1. Continued

component	organism, context	turnover time	Reference
		^a , half-life; ^b , turnover; ^c , lifespan	
finger nail	<i>Homo sapiens</i>	3.47 mm/month	Yaemsiri et al. ¹¹⁰
eye	<i>Schmidtea mediterranea</i>	1–2 months ^b	LoCascio et al. and Cross et al. ^{7,10}
epidermis	<i>Schmidtea mediterranea</i>	5–6 days ^a	van Wolfswinkel et al. ¹¹¹
epithelial cells	<i>Hydra attenuata</i>	4–5 days ^c	Otto and Campbell ¹¹²
sponge	<i>Halisarca caerulea</i>	35%–40% body carbon/day	De Goeij et al. ¹¹³

^aHalf-life values, such as from labeling experiments.

^bTurnover time can refer to a component group, or a component.

^cTurnover time for a component group could differ from average component lifespan if loss is stochastic. The related concept of average component lifespan is therefore used for some cases.

Turnover on the organismal level

Turnover of all cells occurs in planarians and hydra. Why? With the eye example from before in mind, it is easy to envision how an unlimited array of small injuries or deviations in cell-type proportions can be resolved with turnover, providing immense simplification to the problems of recovering from unpredictable injuries, variation, or errors and to regeneration specificity (Figure 5A). Planarians can also grow or shrink with cell-number changes based on nutrient influence on turnover parameters.⁸ Planarian fragments reduce remaining tissues during regeneration (morphallaxis), with smaller states readily achieved by scaling tissue production rates to the new fragment size combined with turnover.⁸ In principle, specific regulation in regeneration could also be added to turnover.¹¹⁴ Hydra present a second whole-body turnover example, and these animals also display whole-body regeneration (Figure 5B). Two proliferative myoepithelial layers (cell division takes four days) continuously progress toward body ends where they are sloughed off; cell loss also occurs through budding.^{115,116} Hydra have multipotent stem cells (i-cells) that continually generate neurons, nematocytes (stinging cells), secretory cells, and gametes; i-cells and their descendants also progress toward hydra ends and are sloughed off.¹¹⁶ Similarly, nematocytes (stinging cells) are continually generated from stem cells in a cellular conveyor belt in the jellyfish *C. hemisphaerica*.¹¹⁷ Small-scale injuries could regenerate in hydra through turnover, and changes in production and loss with food availability can lead to growth or shrinking, similar to planarians.¹¹⁵

Turnover of large biological structures

Bone

If anything is stable and inert in our bodies, it would be the skeleton, right? Wrong. Bone undergoes continuous turnover with degradation by specialized macrophage-like osteoclasts¹¹⁸ and production by specialized fibroblast-like osteoblasts¹¹⁹ (Figure 5C). Ducey et al. (2000) stated that “[b]one is removed, not because it is no longer needed ... but because for some reason it has to be replaced.”¹²⁰ Like digging ditches and filling them back in, bone is continually turned over. Such turnover could arguably serve as preventative maintenance, clearing random fit and deteriorated regions. Osteoblast ablation leads to progressive bone loss in mouse with bone resorption continuing unabated.¹²¹ Osteoporosis can be a result of bone

resorption outpacing production¹²²; with aging, the balance can favor degradation. An estimated 10% of the skeleton is turned over per year,^{106,123} with the actual turnover rate varying across the skeleton. Bone damage results in targeted repair, using the components involved in turnover. The cells involved in bone turnover themselves undergo turnover: osteoclasts live 16 days (measured from labeled precursor nuclei, which fuse in multinucleated cells),¹²⁴ and osteoblasts live for weeks.¹²³

Hair

Have you ever run your fingers through your hair and wondered why perfectly good hairs fall out? Hair undergoes programmatic periods of growth and shedding.¹²⁵ In many species, such as the arctic fox, the hair cycle is seasonal and associated with changing habitat. Some species show synchronous or wave-like hair cycles that are not clearly season coordinated, such as mice. In some species, such as humans and guinea pigs, hair shows asynchronous, mosaic cycles. The hair-follicle cycle includes growth (anagen), regression (catagen), and resting (telogen) phases before growth of a new hair and shedding occur.¹²⁵ (Figure 5D). In many species, hair spends extended time in telogen; in the human scalp, by contrast, the majority of time is spent in growth.¹²⁶ In the destructive phase, cell death degenerates the follicle.¹²⁵ Observations of human hair cycles indicate some hairs cycle over 15 to 34 weeks (e.g., arm and hand) and with an average from 11 to 38 months for the scalp (the scalp growth period is often stated to range from 2 to 6 years¹²⁷).

The evolutionary basis for the hair-follicle cycle has been unclear. Paus (1998) states that “[i]t remains mysterious why the HF engages in the—seemingly wasteful—behaviour of initiating and rhythmically interrupting its growth cycle by entry into catagen and telogen.”¹²⁸ Hypotheses include elimination of harmful chemicals or cells that could be a cancer risk, parasite elimination, length control, cleansing the body surface, or that cycling is associated with retaining some state capable of regeneration upon hair loss.^{129,130} Damage to hair (e.g., clipping) does not trigger growth or replacement.¹³¹ However, hair damage need not be detected for maintenance of a healthy state, because individual hairs will eventually be shed and replaced with new ones regardless of damage. Thus, turnover can maintain a state that does not accumulate damage at the price of also eliminating good hairs. Full hair removal can shorten telogen, and the follicle can enter anagen sooner.¹²⁵ In mice, large skin injuries can be

associated with *de novo* follicle regeneration.¹³² Thus, hair regeneration can involve injury-response-specific processes. However, at the core of hair maintenance is an enduring programmatic cycle of loss and growth—turnover.

Feathers

Birds turn over feathers throughout their lifetime through molting¹⁰⁸ (Figure 5E). For many this occurs annually, whereas others have two to three molts per year. Damage to feathers includes breaks (e.g., after predator engagement), color fading from sun exposure, and wear and tear from normal use. Heavily damaged feathers are evolutionarily costly, and it has long been viewed that molting facilitates the replacement of damaged feathers.¹³³ Annual molting discards many functional feathers as a programmatic process; however, this also accomplishes the removal and replacement of damage without needing to know which feathers are damaged. Feather turnover might seem wasteful given that feathers can be 4%–12% of a bird's weight, yet it will yield the benefits of maintenance of an adult state without damage accumulation and regeneration.

Teeth, chaetae, claws, and nails

Teeth and annelid chaetae present pipeline turnover examples. Some species replace teeth only once (diphyodonts) like humans; however, most replace teeth continuously (polyphyodonts), including in sharks, geckos, alligators, snakes, many amphibians, fish, and dinosaurs.¹³⁴ Polyphyodonty can result in impressive assembly-line-like tooth turnover in sharks (Figure 5F). In leopard and nurse sharks, for example, teeth rows can be replaced at a stunning rate of every 9–12 days and 9–21 days, respectively.^{104,135} Turnover ensures that shark teeth of the front row are continually mature.¹³⁵ Similarly, crocodilians can replace their teeth 50 or more times in a lifespan (approximately once per year), which is programmatic rather than damage triggered.¹³⁶ Triceratops and *Allosaurus* replaced teeth every ~83 and ~100 days, respectively.^{105,137} Loss of continual tooth renewal in mammals is hypothesized to be related to fitting tooth morphologies for chewing (occlusion).¹³⁸ For certain teeth, such as in mice or beavers, continuous growth balances wear and damage as a form of turnover.¹³⁹ Tooth renewal could have evolved to handle frequent damage and/or for enabling growth or morphology change in life history. Regardless, its programmatic nature where it occurs, as beautifully illustrated by shark teeth, can enable regeneration and continual maintenance of a fit, mature state despite damage.

Chaetae are chitinous bristles in annelid worms.⁵⁹ Chaetae turnover also appears to be pipeline-like, with new chaetae forming in one location and then maturing and progressing across a row, with older chaetae either being shed or degraded distal to the location of formation (Figure 5G). As exposed structures, like teeth, chaetae will be subject to damage, and programmatic turnover will enable regeneration and maintenance of a continually fit state.

Nails or claws display continuous growth, in this case balanced by wear; e.g., the American badger completely replaces its claws in an estimated 130–140 days¹⁰⁹ (Figure 5H). Nail and claw damage will be repaired without needing to know anything has gone wrong by this continuous production and extrinsic degradation process.

Antlers, leaves, and needles

Every year, various species of the Cervidae family, e.g., deer, moose, and caribou, shed (cast) their antlers and grow new ones (Figure 5I). Failure to cast an antler still results in regeneration of another antler the following season,¹⁴⁰ suggesting regeneration is programmatic rather than triggered by antler absence. Antler regeneration can provide replacements for antlers damaged during fighting and also enable yearly seasonal growth.¹⁰⁷ Many trees shed their leaves in the fall in a process called abscission (Figure 5J). Myriad conditions can regulate abscission, including damage.¹⁴¹ Many evergreens also shed their needles every one to five years depending on the species.¹⁴² Regardless of regulatory features and evolutionary origins, when programmatic replacement occurs, turnover will renew components, replacing damaged ones.

Turnover of cells within tissues

Varied turnover of adult mammalian tissues

Tissues with extensive turnover are highlighted in the following sections. Other tissues, like the trachea and olfactory bulb in some mammals, have moderate turnover (Table 1). For some tissues with little or no turnover, such as muscle, liver, or blood vessels, damage can trigger proliferation. For some tissues without turnover, such as the central nervous system, cell loss can be permanent. Nonetheless, cellular components in such cells are extensively turned over as described in the sections on organelles and molecules.

Intestine, esophagus, stomach, and colon

The intestinal epithelium is organized into units of crypt (invagination) and villus (protrusion) and is maintained by a burning engine of pipeline turnover (Figure 5K). Stem cells are maintained through neutral competition in the crypt base¹⁴³ and produce a conveyor belt of different cell types that migrate toward villus tips, where they are shed. Small-intestine turnover in mice, rats, and humans takes mere days (Table 1). Similarly, the rat stomach epithelium turns over in about 3 days,⁹⁰ the mouse colon turns over in 1–5 days,^{144,145} the human colon turns over in 3–4 days,⁸¹ and the human rectal epithelium turns over in 6–8 days.⁹¹ The esophagus is maintained from a proliferative basal layer,¹⁴⁶ with a turnover time in humans of roughly 9 days.⁸⁹ Why such dramatic and rapid turnover occurs generally receives less focus than underlying mechanisms. For example, Clevers (2013) states, “there are many outstanding questions ... Why does the intestinal epithelium self-renew with such an unprecedented rate?”¹⁴⁷ Some hypothesized that epithelial flux is related to digestive and absorptive capacity¹⁴⁸ and handling frictional cell loss. The intestine is frequently exposed to cytotoxic compounds, infectious agents, and mechanical damage. A common view is that turnover minimizes epithelium exposure to the harmful environment of the intestine lumen.¹⁴⁹ LeBlond considered that “ordinary repair mechanisms cannot efficiently maintain tissular integrity” and that a “constant supply of young cells anticipates damage and prevents the occurrence of weakened areas or gaps.”¹

Turnover properties described previously (Figures 1, 2, and 3) could be applied to the intestine. Small-scale sporadic damage could regenerate without needing a response, and should a damaging agent kill a particular cell class, regeneration specificity could occur. Similarly, cells with sporadic chemical/

infection damage will be cleared without necessitating damage detection. With the intestine continuously exposed to a toxic environment, cellular fitness should progressively decline. The high turnover rate, however, minimizes the steady-state level of such damage. Rapid turnover also should minimize production delay for repair. Turnover also predicts the robustness of the intestinal epithelium to internal variation in the production ratios of different cell types. After any transient stimulus that changed cell-type ratios, turnover would return ratios to normal. After major injury, additional repair responses are overlaid onto turnover, including inducible stem-cell identity in Lgr5+ descendants and crypt fission.¹⁵⁰ Imperfections in this induced repair process could ultimately be pruned by turnover.

Blood

RBCs represent a massive pipeline turnover engine, with cells living about 120 days and roughly 250 billion RBCs produced per person daily⁸⁵ (Figure 5L). RBCs in mammals extrude nuclei, precluding long life. Nonetheless, RBCs turn over similarly in vertebrates that lack nucleus extrusion, such as in ducks (~40 days lifespan) and chickens (28–35 days lifespan).¹⁵¹ As RBCs age, they are recognized by phagocytic cells and removed. In response to hypoxia, erythropoietin (EPO) increases RBC production by promoting erythroblast progenitor survival (colony-forming unit-erythroid [CFU-E] cells).¹⁵² Therefore, underlying RBC turnover is an even bigger engine of erythroblast turnover, with an excess continually produced and degraded in the spleen. For example, ~40%–60% of mouse erythroblasts undergo apoptosis without making an RBC.¹⁵² Other blood cell types, from dendritic cells (DCs) to regulatory T cells turnover with different timescales (Table 1); at least some, such as DCs, show apparent first-order kinetics consistent with cell age-independent turnover.⁹⁷

Epidermis, sebaceous glands, and cornea

Our skin is continually renewed from basal keratinocytes.¹²⁵ Basal progenitors are maintained by balanced stochastic renewal versus differentiation, and their progeny transit upward and differentiate into the outer skin layer (the stratum corneum).¹⁵³ Production is matched by continual destruction in a pipeline turnover system involving a specialized cell-death process during corneocyte maturation and shedding (desquamation)¹²⁵ (Figure 5M). Radiolabeling and ultraviolet (UV)-light-induced pigmentation loss rates indicate a human epidermal turnover time of ~36–39 days.^{82,154} From stratum corneum turnover observations, the entire epidermis was estimated to turnover in ~45 days.⁸³ Mouse is even more impressive, with transit time from basal through surface corneocyte taking only 8 days.¹⁵⁵ Skin thus turns over continually and rapidly; strikingly, the entire human epidermis is turned over multiple times a year. In terrestrial life, desiccation is a major challenge solved with barrier formation (e.g., cornification). Despite not undergoing cornification, however, zebrafish skin also demonstrates extensive turnover: cells spend 8.4 days on average on the surface, and the entire surface population turns over in 19.5 days.⁸⁴ Many cell-addition events in zebrafish live tracking were remote from cell-loss locations, consistent with turnover largely being random rather than targeted to replace dysfunctional cells. Sebaceous glands in human skin hair follicles turn over dramatically as well, in 7.4 days.⁹⁴

Some suggestions for the purpose of skin turnover involved distance from supporting nutrient-delivery and waste-removal processes and exposure leading to inevitable cell exhaustion and decline.¹⁵⁶ However, preemptive epidermal death in terrestrial vertebrates acts more as a conveyor belt preventing decline from occurring in the first place. Skin is subjected to continuous environmental perturbation, such as dehydration, UV light, temperature fluctuation, friction, infectious agents, chemicals, and mechanical damage. Continual turnover should thus yield the benefits described in the sections on turnover properties for addressing these challenges. Like many turnover systems, major perturbation can also trigger specific regenerative processes, involving migratory and proliferative responses that interface with turnover.^{84,125}

Light enters our eyes through another rapidly turning over and highly regenerative layer of cells called the corneal epithelium. Cell division is balanced by shedding (desquamation), with programmatic turnover taking approximately a week⁸⁶ (Figure 5N). Exposure to the environment will subject cornea cells to damage, arguably necessitating the high turnover rate, enabling regenerative repair and damage clearance.

Taste buds

Most of us probably have not realized that the entire epithelium of our tongues, including taste-receptive cells, is turned over every few weeks (Figure 5O). Taste receptor cells have a lifespan of about 10 days,⁸⁷ and the surrounding keratinized epithelial cells turn over in 2–4 days,⁸⁸ being renewed from bipotent stem cells.¹⁵⁷ Loss of taste receptors shows exponential decay behavior, consistent with stochastic death, although Beidler et al.⁸⁷ interpret this to reflect random exposure to damage. However, programmatic turnover could allow local regeneration and clear damaged cells (at the price of removing non-damaged ones), without needing sensitive cell-fitness surveillance.

Sponge choanocytes and ctenophore tentacles

Turnover is understudied across non-model animals, and impressive cases will undoubtedly be pervasive. For example, a staggering 35%–40% of body carbon is replaced per day in the sponge *Halisarca caerulea*¹¹³ (Figure 5P), and ctenophores (comb jellies) have continuous *piwi*+ stem-cell-derived progenitor streams that are examples of cellular conveyor belts¹⁵⁸ (Figure 5Q). Many sponges and ctenophores are highly regenerative.

Turnover of organelles

Autophagy and proteasomal degradation

In the following sections, autophagy and the proteasome have prominent roles. In macroautophagy (for simplicity, autophagy), cytoplasmic proteins and organelles are surrounded by a double-membrane (the autophagosome), and the contents are destroyed in the lysosome (Figure 5R). In bulk (non-selective) autophagy, heterogeneous contents (cytoplasm and organelles) are degraded, and in selective autophagy, specific components (e.g., mitochondria, lysosomes, endoplasmic reticulum [ER], ribosomes) are targeted for degradation.^{52,159} Autophagy is active in basal conditions for turnover and is induced under nutrient starvation.¹⁶⁰ Autophagy can also mediate some surveillance-based degradation of damaged cellular components, such as mitochondria or lysosomes.¹⁵⁹ The ATP-dependent ubiquitin-proteasome system (UPS) provides another major turnover

engine. In the UPS, ubiquitin is added to proteins through the action of >600 E3 ubiquitin ligases, and ubiquitylated proteins are targeted to the proteasome for degradation¹⁶¹ (Figure 5S).

Eukaryotic flagella

Eukaryotic cilia and flagella display constitutive turnover.^{19,162} The single-celled alga *Chlamydomonas* has two flagella of equal length. Blocking intraflagellar transport (IFT) of new flagella components leads to flagella shrinking and disappearance within hours.¹⁶³ Thus, turnover is impressively rapid, with a 0.011 micron/second estimated disassembly rate. Decreasing, but not eliminating, new component incorporation leads to short flagella, and decreasing disassembly leads to longer flagella, indicating that flagella are dynamic structures in continual flux⁷⁹ (Figure 5T). Cilia can be removed, for example by fluid shearing or laser amputation. Regeneration occurs in approximately ~2 h.¹⁹ If both flagella are removed in the presence of cycloheximide, preventing *de novo* component synthesis, then regeneration still initiates and reaches about half of the initial length.¹⁹ Furthermore, regeneration initiates with essentially no delay. These observations indicate that progenitors for the flagella are available at the time of injury, potentially en route as part of turnover without necessitating damage detection for new progenitor synthesis, consistent with properties yielded by turnover. Flagella regeneration and length control are therefore associated with a constitutive and rapid turnover process.⁷⁹ In this turnover system, several properties impact steady-state length and regeneration dynamics. The disassembly rate at the flagella tip is length independent,⁷⁹ but the assembly through IFT is length dependent, scaling as 1/length, for not fully resolved reasons,¹⁶⁴ but possibly involving change in IFT particle entry rate.¹⁶⁵ There also appears to be transcriptional regulation after flagella removal, added to turnover.¹⁶⁶

Prokaryotic flagella

Prokaryotic flagella are comprised of a basal body, a proximal hook, and a polymerized flagellin filament with a cap at the end. *E. coli* flagella continually grow in a length-independent manner, and broken filaments can continue to grow at this constant rate.¹⁶⁷ The cap protein FliD is continually synthesized and exported, presumably enabling re-capping of a broken flagellum and its regeneration without needing to sense damage and induce synthesis.¹⁶⁸ Constant production and growth of components are hallmarks of a turnover system, with turnover enabling repair, at least for some species. As for degradation, it is hypothesized that long filaments break.¹⁶⁷ Whereas there might not be surveillance on the integrity of any given flagella in at least some species, there is substantial regulation of flagella production by integrating numerous cues to commit to flagella synthesis and during production to monitor the abundance of precursors.¹⁶⁹

Mitochondria

In the yeast *S. pombe*, similarly sized cells produced similar amounts of mitochondria in the cell cycle regardless of their starting amount.¹⁷⁰ In other words, cells that deviate from the mean do not display a compensatory change in mitochondria production. Distribution of mitochondrial volume combines with spacing out of nucleoids within mitochondria to partition mitochondria proportionally to the volume of daughter cells.¹⁷⁰ Cells, given their consistent size in the population, have a mechanism tying cell size to cell division,¹⁷¹ such that any cell size

variation could result in more or less time of constant production, accelerating return to the mean through passive constant production and division. These findings point to non-surveillance core mechanisms for determining mitochondria numbers in this context.

Metabolic labeling in rat liver measured turnover of cytochrome c, soluble and insoluble mitochondrial proteins, and mitochondrial lipid.⁷⁸ All components had a similar half-life of around 10.3 days and displayed first-order decay. These observations suggest that mitochondria are rapidly and continually turned over, are degraded largely as complete organelles, and that turnover to a first approximation is independent of mitochondrial age (Figure 5R). Similar mitochondrial turnover behavior, including first-order kinetics, was observed across rat tissues and ages.¹⁷² There is, however, added complexity. First, mitochondria undergo fusion and fissioning, with fissioning required for mitophagy.¹⁷³ Furthermore, some outer membrane proteins are degraded via the proteasome, and mitochondrial proteins can be degraded within the mitochondrial matrix by the Lon protease and other AAA+ proteases, giving some proteins different turnover rates.¹⁷⁴ Bulk measurements on degradation likely more accurately reflect the behavior of inner mitochondrial membrane proteins than all mitochondrial proteins. Regulation of mitochondrial quality is a major focus as well. For example, outer mitochondrial membrane (OMM) potential disruption can promote mitophagy through PINK1/Parkin on the OMM.¹⁷⁵ Mitophagy can also be upregulated in starvation conditions.¹⁵⁹ Measurements of basal mitophagy rates across mammalian tissues show that it is pervasive *in vivo* and variable in level across cells.¹⁷⁶ Study across cell types and organisms will be important for dissecting the relative contributions of generic mitochondria turnover versus damage-triggered turnover.

Golgi, vacuole, and endoplasmic reticulum

Some organelles are in a constant state of flux. For instance, Golgi is a structure of components in flux with abundance determined by balance between continual production and loss through cisternae maturation and cell division^{177,178} (Figure 5U). Such turnover should enable maintenance in a fit state, return from deviation, and repair. Statistical analyses in yeast suggest Golgi and vacuole abundance are not overtly under surveillance-like control, indicating that organelle abundance is not tightly regulated in at least some systems.¹⁷⁹ Experiments that perturbed vacuole inheritance indicated that vacuole growth rate was not regulated by greater vacuole abundance in mother yeast cells or lower vacuole abundance in daughters, pointing to a passive process for determining abundance.¹⁸⁰

The flux for at least some organelles can be regulated by assessing their function. The ER is the site of folding and modification of secreted and integral membrane proteins, with continual protein flux. ER membrane is turned over with a half-life of ~2 days, and labeled components show exponential decay.^{80,181} ER turnover can occur through selective autophagy (ER-phagy); blocking ER-phagy with inhibition of FAM134b leads to ER expansion in culture and in knockout mice.¹⁸² ER-phagy thus appears to impact constitutive ER turnover and is also upregulated under nitrogen-starvation conditions or rapamycin treatment.¹⁸³ Production of ER (including resident proteins and

lipids) is regulated, associated with the detection of unfolded proteins in the ER through the unfolded protein response (UPR), setting ER levels by trafficking demands.¹⁸⁴

Turnover and macromolecular structures

Macromolecular complexes

Components of macromolecular complexes can also be degraded, although sometimes at lower rates than average for the proteome. Striated muscle involves the large and seemingly stable sarcomere, comprised of myosin, actin, and titin filaments. Titin is the largest protein encoded in many genomes at ~3.8 million Daltons. Surely the sarcomere, essential for muscle contraction, is stable? Apparently not. Titin molecules have a half-life of 3–5 days in cultured muscle,⁷⁷ and replacement of titin in the sarcomere can occur even faster (half-life of ~2.1 h).¹⁸⁵

Synapses

In mammals, most neurons are not replaced, with the exception of the olfactory bulb in some species¹⁸⁶ and the hippocampus.¹⁸⁷ Thus, neurons can live for 100+ years. Despite this remarkable capacity for long life, neuron components continually turn over. Synapses were long envisioned, like the neuron itself, to be static. Not so: the components of synapses are quite labile. The long-lasting nature of memories gives an illusion of stability, with the molecular substrate of our thoughts, instead, rapidly turning over.¹⁸⁸

One approach to assessing synapse stability involves imaging dendritic spines. About 60% of spines in the mouse neocortex are stable, where they could be involved in memory maintenance.¹⁸⁹ By contrast, in the mouse hippocampus, dendritic spines are strikingly short lived, with a mean lifetime of 1-to-2 weeks,¹⁹⁰ possibly entirely turning over in weeks.¹⁹¹ Even when spines are stable, however, their components need not be. For example, using a photoactivatable green fluorescent protein (GFP), post-synaptic scaffold protein PSD-95 was found to be dynamic, with a retention time of only ~100 min.¹⁹² Imaging of synapses also shows they can dynamically change size.¹⁹³

Neurotransmitter receptors are dynamic in membranes, showing diffusion, endocytic recycling, and degradation.¹⁹⁴ Scaffolding molecules that maintain synapse structure are also dynamic. The half-lives of numerous synaptic proteins were typically 2-to-5 days in cell culture.^{39,195,196} Proteins in the mouse brain analyzed *in vivo* had an average lifetime of 9 days.⁶⁶ In the mouse cortex, most proteins had lifetimes of 3–13 days, although some proteins could last for several hundred days.⁶⁸ In rat primary cortical neuron cultures, synaptic proteins had half-lives of ~2–5 days, including post-synaptic scaffold proteins such as Shank2 or PSD-95.¹⁹⁶ Similarly, the mean half-life of 184 synapse proteins in primary hippocampal cultures was roughly 5 days.³⁹ This included post-synaptic scaffold proteins such as PSD-93, PSD-95, and SAP-97; adhesion molecules such as neuroligins, neuroligins, and N-cadherin; neurotransmitter receptors such as GluR1; and presynaptic proteins such as Unc13a. Some synaptic proteins had half-lives of 10 or more days. Much of the mammalian brain proteome is thus estimated to turn over in days to weeks—a paltry time frame compared to the decades neurons can live.

Degradation can involve the UPS system, autophagy, and the endolysosomal pathway¹⁹⁷ (Figure 5V). Synthesis and degrada-

tion of key components can be regulated, such as by activity, and turnover can enable dynamic change, such as long-term potentiation.¹⁹⁸ Constitutive turnover of synapse components should also yield other desirable properties, including clearance of variable and constitutively occurring damage—maintaining a state with low basal damage levels, rapid repair, and low hazard formation risk.

Vertebrate photoreceptors

Photoreceptor neurons of vertebrate eyes are in continual, rapid flux. The outer segment of rods is comprised of stacked disc-like membranes containing light-detection proteins, prominently opsin, within an outer membrane. Radioactive pulse labeling showed continuous progressive movement of radiolabeled proteins through outer cell segments until being lost¹⁹⁹ (Figure 5W). Similar observations were made in frogs, mice, rats, and rhesus; e.g., labeled proteins were completely lost by ~9 days in rats.^{74,200} The membrane discs of rods are continually assembled at the base of the outer segment, progress distally, and then are shed from the cell tip and phagocytosed by the retinal pigmented epithelium (RPE).²⁰¹ This spectacular example of intracellular pipeline turnover is reminiscent of some cellular systems, like the intestine. Young noted the extreme metabolic investment in photoreceptor renewal and described this in the context of investment in general renewal, such as for RBCs and intestines.²⁰² Each day it is estimated that a photoreceptor will synthesize 80–90 discs and that an RPE cell phagocytoses 25,000–30,000 rod discs. Cones similarly turn over components.²⁰² Exposure to light is highly damaging, resulting in reactive oxygen species and damage; turnover can help remove this damage without having to detect and specifically repair damaged components. Our photoreceptor neurons should thus experience a continually fit state through rapid turnover despite constitutively damaging conditions.

Ribosomes

Ribosomes make up an estimated 40% of the yeast cytosol²⁰³ and ~6% of the protein and ~80% of total RNA in mammalian cells. Ribosomes can be degraded by autophagy (Figure 5X); for example, being visible in lysosomes.²⁰⁴ Metabolic labeling in rat and mouse liver suggests that the majority of ribosomal proteins and rRNA are degraded with similar, and first order, kinetics, consistent with entire ribosomes being degraded.^{73,205} The first order kinetics suggest to a first approximation that degradation is random rather than targeted to old ribosomes. Some particular ribosomal proteins are degraded at significantly different rates, consistent with some exchange with a free pool.²⁰⁵ The *in vivo* data suggests a ribosome turnover time of ~7 days.⁷³ Similarly, ribosome half-life in *Arabidopsis* cell culture was ~3–4 days.²⁰⁶ Hirsch (1966) noted, “... the physiological significance of this continuous replacement ... has remained obscure”. Turnover properties predict that damaged or dysfunctional ribosomes will be replaced and prevented from accumulating, while paying the price of degrading functional ribosomes as well. Similar to other turnover systems, ribosome turnover can also be regulated. In conditions of stress, such as following nitrogen or amino acid starvation, ribosome degradation occurs rapidly.²⁰⁷ Nonfunctional rRNA decay (NRD) can eliminate rRNA from dysfunctional ribosomes.²⁰⁸ Finally, translation is

under surveillance; stalling can cause ribosome collisions and trigger ribosome-associated quality control (RQC), with the mRNA and nascent polypeptide degraded.²⁰⁹

Bacterial cell wall

Bacterial cell walls are defensive barriers and also display dramatic turnover: roughly 50% of the cell wall turns over per bacterial generation, including in gram-negative and gram-positive species^{75,76,210} (Figure 5Y). Turned-over components can be recycled in the cytoplasm, especially in gram-positive species. For example, the vast majority of murein peptidoglycan (PG) comprising the *E. coli* cell wall is recycled through fragment reuse in the cytoplasm. In gram-positive bacteria, large cell-wall fractions are shed into the media in the form of muropeptides. Because the cell wall is massive (~20% of the weight of a gram-positive bacterium and 3% of a gram-negative bacterium), turnover is a major investment. Turnover occurs as a first-order reaction indicating that it appears random.⁷⁵ Why such extensive turnover of the cell wall occurs is often noted to be unclear. In the article “Why does *Escherichia coli* recycle its cell wall peptides”, Park states, “... there is no apparent need for wall turnover, we are left with the possibility that recycling serves some signaling or regulatory role(s)”.²¹¹ Common hypotheses include that released muropeptides mediate communication, that turnover can be used to synthesize beta-lactamase in response to damage induced by antibiotics, or that it mediates aspects of virulence and resistance.^{212,213} As an exterior layer, the wall is exposed to harsh conditions including pH, lysozymes, antibiotics, mechanical stress, and chemical agents.²¹⁴ Breaching cell-wall integrity can mean death. Some factors are involved in response to damage,²¹⁵ and cell-wall constituents can also be modified to enable growth and protection.²¹⁶ However, massive baseline turnover should yield many benefits, with a fit state maintained despite fluctuations and continual and sporadic damage.

Turnover of molecules within cells mRNA

mRNAs are subjected to shockingly rapid degradation, often shortly after being synthesized. Constant mRNA turnover throughout the transcriptome was observed following metabolic labeling.^{217,218} In mouse fibroblast 3T3 cells, mRNA had a strikingly short median half-life across genes of 2.1 h.⁶⁴ There are substantial differences in the decay rate of different transcripts; some long-lived mRNAs, such as those encoding ribosomal proteins, can persist for a day or more. The global, massive degradation of mRNA cannot readily be explained by targeted removal of dysfunctional or damaged RNAs; specialized mRNA destruction complexes function to destroy perfectly functional mRNA. Deadenylation typically precedes mRNA degradation²¹⁹ and is progressive.²²⁰ Deadenylation is an active process catalyzed by the CCR4-NOT and the PAN2/PAN3⁶³ complexes and leads to transcript degradation by the RNA exosome, a 3' to 5' exoribonuclease²²¹ (Figure 5Z). Alternatively, 5' caps on deadenylated mRNAs are removed by the Dcp1-Dcp2 complex, allowing message destruction by the 5' to 3' XRN1 exoribonuclease.⁶³ MicroRNAs (miRNAs) and tRNAs last longer than mRNAs, but these RNAs are turned over as well, in the timescale of days (Table 1).

What sets the decay rate of different otherwise functional mRNAs is an active area of investigation. Translation efficiency, such as is affected by codon usage or poly(A) tail length, is a proposed determinant.^{63,222} mRNA lifetime and its poly(A) tail length are correlated, pointing to deadenylation rate as a major determinant of time to degradation.⁶⁴ There are additional sequence-specific regulatory mechanisms that can influence deadenylation and decay rates, such as targeting by miRNAs or particular RNA-binding proteins.^{63,223} There are also some surveillance mechanisms to facilitate degradation of certain dysfunctional mRNAs. For example, in nonsense-mediated decay, mRNAs with premature stop codons are recognized and degraded.²²⁴ In No-go decay, mRNAs associated with a stalled ribosome are degraded.²²⁵

Global mRNA turnover should facilitate maintenance of a stable transcriptome state despite fluctuations and cellular or mRNA damage, maintain a constitutively low level of damaged transcripts, remove transcription errors, lower RNA aggregate hazard formation risk, and facilitate regulation and state transitions (e.g., during differentiation or stress). For example, fast mRNA degradation enables rapid reaching of a steady state for some stress response genes.²²⁶ Desirable properties for encoded protein levels or dynamics should also influence mRNA turnover evolution.

Proteins

Protein levels result from mRNA and protein production and degradation rates, and, in dividing cells, from dilution. Peptide bonds hydrolyze without active proteolysis with half-lives in the 100s of years range.⁵³ However, the proteomes of cells, whether dividing or not, are in a continual state of rapid turnover, requiring extensive energetic investment. For example, translation is responsible for an estimated 30%–75% of energy expenditure in diverse cells, with synthesis of an average polypeptide costing ~3,000 ATPs.²²⁷ ~40% of rat liver proteins are turned over per day²²⁸ and ~4 million proteins are degraded per hour in a yeast cell.³⁸ Even non-dividing cells, ranging from bacteria,³⁵ to carrot,²²⁹ to yeast,²³⁰ to mammalian cells,²³¹ display extensive protein turnover. Mass spectrometry and isotope labeling has given protein turnover studies incredible resolution.^{65,195,217} For example, in non-dividing mouse C2C12 myoblasts, most of the proteome turned over within ~2 days.⁶⁵ *In vivo* measurements indicated average lifetimes of 9, 3, and 3.5 days for proteins in the brain, liver, and blood, respectively.⁶⁶

Proteins are degraded by the UPS, the lysosome-autophagy system, and through mitochondrial proteases. Genetic experiments blocking autophagy indicated that in basal conditions, short-lived proteins are predominantly degraded by ATG5/7-independent mechanism (e.g., UPS) and that a large fraction of long-lived proteins are degraded by both ATG5/7-autophagy-dependent and independent mechanisms.²³² Thus, autophagy and UPS act in concert for turnover of the proteome in basal conditions.

As described above, the evidence on balance indicates that the vast majority of molecular biology operates without surveillance of protein levels or age. There are, naturally, cases of feedback and mechanisms to detect certain kinds of protein aberrations, such as folding errors, sometimes referred to as quality control (QC). For example, in ER-associated protein degradation

(ERAD), misfolded ER proteins are detected and delivered to the cytosolic proteasome.²³³ Misfolding ER stress can upregulate the UPR, which increases ER production and transiently lowers new protein production into the ER.²³⁴ Of course, the need for continuous extreme protein flux through the ER is dictated by the short half-life of the proteins transiting the ER. Similarly, misfolded cytosolic proteins can trigger the heat shock response that tunes the ability to handle proteotoxic stress.²³⁵

What is the purpose of the massive and rapid degradation of proteins? Most researchers suggest it is for transitions. Dörrbaum et al. state, “Turnover rates have likely been optimized ... such that individual proteins possess a life time that represents the balance between energy-saving stability and dynamic flexibility.”³⁹ Price et al. also state, “The constant renewal of the protein population is an energy-intensive process, yet it allows the cell to rapidly modulate protein levels in response to changes in the environment.”⁶⁶ Yet in non-dividing cells and for proteins not responding to change, turnover is pervasive nonetheless. The fact that programmatic turnover is the dominant mechanism for degradation suggests that turnover will help yield desired protein levels, enable level stability despite production variation or partitioning error, bring about regeneration, prevent damage accumulation in normal life, clear variable and hard-to-predict damage with specificity, allow recovery from production mistakes, prevent aggregate formation, as well as enable proportion and transition regulation.

Membranes

Plasma membranes are in a continuous state of protein and lipid turnover.²³⁶ Radiolabeling experiments transformed the view of membranes from static to dynamic, turning over entities.^{72,237} Half of plasma membrane phospholipids can be turned over on the scale of one to three days, including in non-dividing cells.^{72,80} Plasma membrane turnover through endocytosis can lead to recycling of components to the membrane or to sorting to the lysosome where components are degraded (Figure 5AA). Some authors speculated that turnover could be important for maintaining membrane integrity, though understanding was poor.^{237,238} For example: “The most likely reason for phospholipid turnover then appears to be that it sustains membrane integrity and membrane function. Such a vague conclusion is not very satisfactory, however, since it in no way explains the necessity for continuous synthesis and degradation of phospholipids.”²³⁷ Whereas there are also responses to membrane damage and disruption,²³⁹ turnover presents a generic mechanism to prevent accumulation of damage (e.g., oxidation of fatty acid chains) and to repair certain damage without surveillance.

TURNOVER AND AGING

From an evolutionary perspective, aging can result in a species from the declining force of natural selection with age.^{240,241} Because individuals die from environmental hazards, late age classes can contribute less to future populations, with selection thus weaker on late-life traits. Furthermore, organisms could evolve to optimize repair investment related to their life-history strategy, such as in trade-offs with reproductive output as in the disposable soma theory.²⁴² There are numerous theories related to mechanistic bases for aging, including wear-and-

tear, epigenetic, immune, endocrine, and damage class accumulation theories, as well as a variety of aging hallmarks,²⁴³ which will not be reviewed here. In prior sections, I described turnover functions and suggested the basis for its levels. Here, how turnover could underlie many contexts of evolved repair investment level (e.g., as in the disposable soma theory) and how the limits on turnover relate to several mechanistic aging models and hallmarks are described. I suggest that such a turnover framework or theory of aging provides context for multiple known aging processes and mechanistic connection with evolutionary theories.

Turnover is fundamental to longevity across systems. Evolutionary optimization could ensure viability over short time scales for some systems (e.g., intestine cell turnover in days), with pleiotropic benefits toward organism longevity (e.g., over decades). The benefits of turnover, such as regeneration, the clearance of damage, and the minimization of constitutive damage levels help promote system and organism longevity. Regulated response systems for repair are naturally important as well. In organisms with proposed agelessness, like planarians and hydra, all cells of the body turn over.

The limits of turnover could contribute fundamentally to aging: (1) *aberrations resistant to turnover*. Such aberrations will occur with chance events or damage accrual, and increase with time. For example, senescent cells can accumulate, inhibit their own removal, and contribute to aging.³¹ Protein aggregates or cross-linking can lead to hard-to-degrade components, such as in Alzheimer’s disease.³⁰ High turnover can lower the formation rate of such aberrations but be insufficient to remove them when they occur. Furthermore, some damage leads to states, such as scarring, that turnover cannot readily resolve, as in wear-and-tear aging models. (2) *Stable epigenetic errors in turnover*. Some errors during turnover could become stably propagated, also leading to defect increase with time. These could be considered epigenetic deviations that are stably maintained. For example, epigenetic clocks show continual progression from a youthful state with age.²⁴⁴ (3) *Constraints: components that do not turn over*. Some components cannot undergo turnover (or undergo slow turnover) because of constraints. First, turnover of some components could be incompatible with system or organism function. For example, neuron turnover could be limited by desired activity-modulated attributes; certain structures (e.g., insect wings, organs) might not be able to be replaced without function loss. DNA is not maintained by turnover, with damage only addressed by surveillance. Mutations and cancer risk increase can thus occur with aging. Second, turnover of some components could be constrained by lack of access to evolved turnover mechanisms. For instance, certain long-lived molecules (e.g., nuclear pore complexes) might be isolated from degradation machinery. It has been argued that such molecular components could display continually increasing damage with age.²⁴⁵ (4) *Evolved low turnover rates*. Previously, I suggested that system turnover rates are based on trade-offs and constraints (Figure 4). Some components might evolve to undergo little turnover; e.g., because of low damage probability. Selection on turnover levels for longer timescale system properties, in many cases, could relate to the life-history strategy of the organism and potentially be limited by the declining force of natural

selection with age and trade-off costs toward reproduction. Turnover levels needed for late-life fitness of some systems simply might not have been the favored outcome of evolution for these reasons, and underlying components could progressively be lost (e.g., certain cells) or deteriorate. Regardless of constraints or low selective pressure, many human cell types in addition to neurons undergo little or no turnover, including cardiomyocytes, hepatocytes, muscle cells, endothelial cells, and kidney cells. Simple wear of components that do not get repaired or replaced (e.g., human teeth) has long been suggested to contribute to aging. (5) *Turnover source competition*. Some turnover systems involve stem cell maintenance through competition, which can remove unfit progenitors.¹⁵³ However, competition also introduces peril, when genetic or epigenetic alterations give a selective advantage and contribute to aging-related changes.²⁴⁶ For example, hematopoietic stem cells become largely monoclonal in late life²⁴⁷ and ultimately, cancer can emerge. (6) *Increasing turnover demand*. Evolved turnover levels might be insufficient to maintain systems once other problems have accumulated (e.g., damage in low-turnover components). Such defects could impact the error/damage rates for other components that would otherwise be kept fit through the evolved turnover rate. (7) *Turnover machinery exhaustion*. Turnover machinery itself can experience fitness decline or become exhausted. Proteome management failures increase with age and autophagic flux can decline during aging.²⁴⁸ Autophagy stimulation can prolong mouse lifespan.²⁴⁹ Some lifespan-extending treatments, such as caloric restriction, metformin, or rapamycin-based mTor inhibition, can enhance autophagy as well as decrease protein production.²⁵⁰ Therefore, defects in proteome turnover are associated with aging and promoting turnover can enhance lifespan and/or healthspan. The turnover capacity of some stem cells can also be limited, with their decline contributing to aging. For example, hair graying occurs when melanocyte stem cells decline.²⁴⁷

Concluding remarks

Biology utilizes energy to maintain steady states. Not all units of organization involve turnover, and the timescale of turnover varies dramatically. This suggests that underlying biological principles might be central to explaining the evolution of turnover and alternatives. Turnover is frequently rapid where damage is constitutively incurred, damage risk is high, or where variation is common. For example, massive investment in turnover occurs in tissues exposed to damaging conditions, like the intestine or tongue, in organisms with high regenerative capacity, such as planarians or hydra, or in components of cells associated with damaging conditions, such as the bacterial cell wall or the membrane discs of photoreceptor neurons. Cellular constituents—from most mRNAs, proteins, and membranes, to complexes and organelles—are rapidly turned over, even in non-dividing states. Long-lived systems such as neurons or synapses or sarcomeres still show substantial turnover of underlying components. Turnover is frequently blind to whether the component to be degraded is old or actually damaged, consequently often degrading functional components. This, in many contexts, has obscured its purpose. A framework, or turnover theory, can help address the purpose and basis for levels of turnover across

systems. Turnover enables regeneration and with specificity, maintenance of states with low constitutive damage levels despite wear and inevitable damage, clearance of sporadic damage, prevention of stable hazard formation, minimal delays in repair time, correction of building errors, state transitions, and regulation of component levels. Exclusively surveillance-based solutions for certain contexts can be limited or inviable, present other challenges to the system, and/or cannot be reasonably achieved in evolution. The alternative to turnover might result in a constant state of partial system disrepair. When regulation is applied, it is frequently though not always, overlaid onto turnover. I suggest that constraints and trade-offs between the metabolic costs of turnover and turnover benefits determine its presence and rate and are tuned by parameters relevant for each particular system, including rate of system wear, exposure level to damage, and need for regulation. Finally, I suggest that the limits of turnover, including damage recalcitrant to turnover, turnover errors, constraints on turnover, systems with low evolved turnover levels, and turnover machinery exhaustion fundamentally contribute to the basis for aging. This framework has broad explanatory ability, such as for predicting system properties and addressing the evolution of turnover levels. Turnover occurs surprisingly rapidly and widely in biology to address numerous challenges and enable life.

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DECLARATION OF INTERESTS

The author declares no competing interests.

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